

NER Smart Logistics & Accessibility Intelligence Platform

User Roles, Role-Based Access Control (RBAC) & Security Audit Specification

This official operational specification documents the 4 authenticated user roles in the NER Smart Logistics Platform, their access credentials, precise feature modification authorities, backend security enforcement (returning HTTP 403 Forbidden on unauthorized calls), and automated verification test results across all 8 North Eastern States.

1. User Role Summary & Login Credentials

Role	Demo Email	Password	Governance Persona & Regional Scope
Admin	admin@nerlogistics.gov.in	admin123	Platform Administrator / Central Logistics Command Director
Government Official	official@nerlogistics.gov.in	official123	State Disaster Authority / Highway PWD Engineer / District Collector
Field Officer	field@nerlogistics.gov.in	field123	On-Ground Highway Patrol / Road Safety & Geotechnical Inspector
Driver	driver@nerlogistics.gov.in	driver123	Essential Goods Carrier Operator (Assigned: Rajesh Kumar / Truck NER-101)

2. Detailed Modification & Operational Rights by User Role

■■■ 1. Admin Role (`admin`)

Primary Duty: Master oversight, security governance, and uninterrupted multi-state supply chain continuity.

What they can Modify / Create:

- **Highway & Road Status:** Full authority to override road statuses (Open, Risky, Blocked) via Live GIS Map with automatic audit logging.
- **Delivery CRUD:** Full authority to create, modify, reassign, reprioritize (Critical/High), and cancel logistics consignments.
- **Emergency Broadcast:** Sole authorized role to broadcast high-priority disaster alerts across all agencies.
- **Incident Final Resolution:** Exclusively authorized to transition confirmed incidents to Resolved.
- **Field Report Promotion:** Convert ground inspection reports into live emergency incidents.
- **System Auditing:** Unrestricted access to governance audit logs (/api/audit-logs) and simulation controls.

■■ 2. Government Official Role (`government\_official`)

Primary Duty: Disaster response coordination, highway hazard mitigation, and corridor contingency planning.

What they can Modify / Create:

- **Road Accessibility Overrides:** Modify highway status to Blocked or Risky based on ground alerts.
- **Incident Progression:** Elevate incidents from Reported to Under Investigation and Confirmed. (Resolving is restricted to Admin).
- **AI Bypass Route Selection:** Analyze blocked corridors and select official alternative detours (e.g. NH-44 Bypass, +42 min) dispatched to drivers.
- **Field Report Approvals:** Convert validated field reports into official active incidents.
- **Restrictions:** Cannot dispatch deliveries, broadcast emergency alerts, or mark incidents as Resolved (returns HTTP 403 Forbidden).

■ 3. Field Officer Role (`field\_officer`)

Primary Duty: Rapid on-ground reconnaissance, GPS geo-tagging, and remote area incident reporting.

What they can Modify / Create:

- **Online & Offline Field Reports:** File hazard reports (Landslide, Flood, Washout) with description and severity.
- **Device Geolocation:** One-click automatic capture of precise GPS latitude and longitude via device sensors.
- **Offline Queue Storage:** Automatically queues inspection reports in browser local storage when disconnected from cellular networks.
- **Batch Synchronization:** Click Sync Now to bulk-upload pending offline reports to MongoDB upon network reconnection.
- **Photo Evidence:** Attach on-site camera photographs of mudslides and debris.
- **Restrictions:** Cannot modify road statuses, dispatch deliveries, confirm/resolve incidents, or broadcast alerts (returns HTTP 403 Forbidden).

■ 4. Driver Role (`driver`)

Primary Duty: Safe point-to-point delivery of critical medical supplies, vaccines, fuel, and relief rations.

What they can Modify / Create:

- **Vehicle Telemetry Console:** Update live speed (km/h), fuel tank level (%), GPS coordinates, and traveling status for their assigned vehicle (NER-101).
- **Strict Vehicle Scoping:** Attempts to update unassigned fleet vehicles (e.g. NER-105) are blocked with HTTP 403 Forbidden.
- **Reroute Acceptance:** Receive detour alerts on console and click [ACCEPT REROUTE] when main corridor is blocked.
- **Alert Acknowledgement:** Read and acknowledge hazard advisories on route.
- **Restrictions:** Simplified interface focused purely on assigned vehicle and deliveries. Cannot create/edit/delete deliveries, modify roads, or access analytics (returns HTTP 403 Forbidden).

3. Feature Authorization Matrix & HTTP Security Enforcement

Feature / Endpoint	HTTP Method	Admin	Govt Official	Field Officer	Driver
Modify Road Status (Open/Blocked)	PUT / PATCH	200 OK	200 OK	403 Forbidden	403 Forbidden
Dispatch Delivery Consignment	POST	201 Created	403 Forbidden	403 Forbidden	403 Forbidden
View Assigned Deliveries	GET	200 (All)	200 (All)	200 (All)	200 (Assigned)
Broadcast Emergency Alert	POST	201 Created	403 Forbidden	403 Forbidden	403 Forbidden
Acknowledge / Read Alert	PUT	200 OK	200 OK	200 OK	200 OK
Report Hazard / Incident	POST	201 Created	201 Created	201 Created	403 Forbidden
Confirm Incident Status	PUT	200 OK	200 OK	403 Forbidden	403 Forbidden
Resolve Incident Status	PUT	200 OK (Sole)	403 Forbidden	403 Forbidden	403 Forbidden
Select AI Alternate Bypass	PUT	200 OK	200 OK	403 Forbidden	403 Forbidden
Accept Reroute on Console	PUT	200 OK	200 OK	403 Forbidden	200 OK
Update Vehicle Telemetry	PUT	200 (All Fleet)	200 (All Fleet)	403 Forbidden	200 (Own Only)
Submit & Batch Sync Reports	POST	201 Created	201 Created	201 Created	403 Forbidden
Deep Analytics & Audit Logs	GET	200 OK	200 OK	403 Forbidden	403 Forbidden

4. Automated RBAC Verification Suite Results

A rigorous automated test suite (backend/test_rbac_suite.js / test_rbac.bat) was executed against an active ephemeral server instance connected to the MongoDB database. The suite verified 36 permission cases with **0 failures**:

Test Scenario	Verified Workflow & Assertion	Security Result
Test 1: Road Status	Admin & Govt can set roads to BLOCKED. Driver & Field calls blocked.	PASSED (200 / 403)
Test 2: Delivery CRUD	Admin creates delivery. Driver views assigned. Non-admin create attempts blocked.	PASSED (201 / 403)
Test 3: Alert Broadcast	Admin broadcasts emergency alert. Driver reads alert. Non-admin broadcast blocked.	PASSED (201 / 403)
Test 4: Incident Resolution	Field reports, Govt confirms. Non-admin attempts to Resolve rejected with 403.	PASSED (Admin Only)
Test 5: AI Reroute	Govt selects alternate bypass (+42 min). Driver accepts on console. Driver cannot set route.	PASSED (200 / 403)
Test 6: Telemetry Scoping	Driver updates own vehicle (NER-101). Driver updating other vehicles rejected with 403.	PASSED (Strict Scope)
Test 7: Offline Sync & Convert	Field Officer syncs batch. Govt converts report to official incident. Driver create blocked.	PASSED (201 / 200 / 403)
Test 8: Full RBAC Matrix	Analytics breakdowns & audit logs accessible to Admin/Govt; 403 for Driver/Field.	PASSED (36/36 Assertions)

5. One-Click Batch Scripts for Windows Operations

- **run.bat**: Boots Python ML (Port 8000), Backend API (Port 5000), and React Frontend (Port 5173), displays credentials, and opens browser automatically.
- **test_rbac.bat**: Executes the automated 36-assertion RBAC security verification suite with one double-click.
- **seed.bat**: Seeds MongoDB with all 10 collections (Districts, Roads, Routes, Vehicles, Incidents, Deliveries, Alerts, Weather).
- **stop_all.bat**: Cleanly locates and terminates all listening processes on ports 8000, 5000, and 5173.